

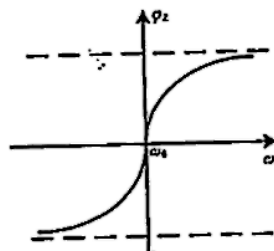
杭州电子科技大学学生考试卷 (A) 卷

考试课程	通信电路与系统	考试日期	2022 年 6 月 22 日	成绩	
课程号	A040095s	教师号		任课教师姓名	陈瑞、林弥、李航、 刘国华、刘艳、张伟、 孙鹏飞、苏江涛、关 阳阳
考生姓名		学号 (8 位)		年级	

题目	第一大题	第二大题	第三大题	第四大题	第五大题	第六大题
得分						

I. Single Choice (3' each)

- Which of the following circuits is not part of AM receiving system? (D)
 - Local Oscillator
 - Detector
 - High Frequency Amplifier
 - Modulator
- A LC resonant circuit has impedance characteristic curve shown as below. When operating at the resonant frequency ω_0 , the resonant impedance of the circuit shows (C).



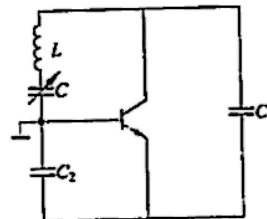
- Capacitive
- Inductive
- Minimum Resistor
- Maximum Resistor

- Consider that the power consumption of power amplifiers is the main energy consumption of wireless communication system. In order to improve efficiency and realize green

communication, which kind of power amplifier is the best choice? (C)

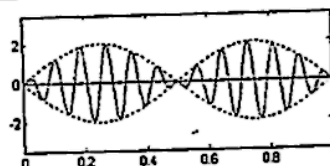
- Class-A
- Class-B
- Class-C
- Class-AB

- If f_c and f_l are the frequencies of input carrier signal and local oscillation signal respectively, then the output intermediate frequency signal is $f_i (=f_l - f_c)$ through a mixer. The frequency component $2f_c - f_l = f_l + 300\text{Hz}$. What kind of interference does it belong to? (A)
 - Whistle Interference
 - Side-channel Interference
 - Cross-modulation Interference
 - Intermodulation Interference
- A feedback LC oscillator is given as below. What kind of circuit is it? (C)



- Colpitts Circuit
- Hartley Circuit
- Clapp Circuit
- Seiler Circuit

- In parallel mode crystal oscillator, the crystal acts as a/an (A).
 - Inductor
 - Capacitor
 - Resistor
 - Varactor
- Which of the following characteristics of diode or transistor cannot be used for amplitude modulation? (D).
 - Base Modulation Characteristic
 - Collector Modulation Characteristic
 - Switch Characteristic
 - Varactor Characteristic
- The time domain waveform of the output modulated signal is shown as below. Which kind of modulation is it? (B).



- A. Standard AM
B. DSB AM
C. SSB AM
D. FM

9. An angle modulated wave is $u(t) = 10\cos(2\pi \times 10^6 t + 5\cos 2\pi \times 10^3 t)$ (v). What is the maximum frequency shift? (C)

- A. 10 KHz
B. 10 MHz
C. 5 KHz
D. 5 MHz

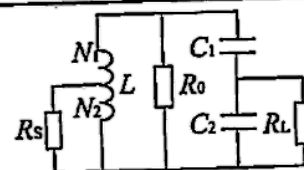
10. For angle modulated wave, which of the following description is wrong? (B)

- A. For standard FM wave, the maximum frequency shift will change only with the amplitude of modulation signal
B. For standard PM wave, the maximum frequency shift will change only with the amplitude of modulation signal
C. For standard FM wave, the maximum phase shift will change both with the amplitude and frequency of modulation signal
D. For standard PM wave, the maximum phase shift will change only with the amplitude of modulation signal

II. Calculation

1. (10') The LC parallel resonant circuit is shown as below, the source resistor R_s and load R_L are partially accessed by auto-transformer and capacitor divider. Known that $L = 1\mu\text{H}$, $Q_0 = 100$, $C_1 = C_2 = 10\text{pF}$, $R_s = 10\text{k}\Omega$, $R_L = 10\text{k}\Omega$. Suppose that R_L and R_s achieve impedance matching.

- (1) Calculate the resonance frequency f_0 and the resonant resistance R_0 (R_L and R_s are negligible).
(2) Calculate N_2/N_1 , loaded quality factor Q_L and bandwidth $B_{0.7}$.



Solution:

$$(1) \quad f_0 = \frac{1}{2\pi\sqrt{LC_2}} \quad C_2 = \frac{C_1 C_2}{C_1 + C_2} = 5\text{pF}$$

$$f_0 = \frac{1}{2\pi\sqrt{10^{-6} \times 5 \times 10^{-12}}} \text{ Hz} = 71.2\text{ MHz} \quad (2')$$

$$R_0 = Q_0 \omega_0 L = 100 \times 2\pi \times 71.2 \times 10^6 \times 10^{-6} \Omega = 44.7\text{ k}\Omega \quad (2'')$$

$$(2) \quad n_2 = \frac{C_1}{C_1 + C_2} = 0.5 \quad R'_L = \frac{R_L}{n_2^2} = 40\text{ k}\Omega$$

$$R_s = R_0 \parallel R'_L = \frac{R_0 R'_L}{R_0 + R'_L} = 21.1\text{ k}\Omega$$

$$n_1 = \frac{N_2}{N_1} \quad R'_s = \frac{R_s}{n_1^2} \quad R'_s = R_s \quad n_1 = \sqrt{\frac{R_s}{R'_s}} = \sqrt{\frac{10}{21.1}} = 0.69 \quad (2''')$$

$$Q_L = \frac{R'_s \parallel R'_L}{\omega_0 L} = \frac{0.5 \times 21.1 \times 10^3}{2\pi \times 71.2 \times 10^6 \times 10^{-6}} = 23.6 \quad (2''')$$

$$B_{0.7} = \frac{f_0}{Q_L} = \frac{71.2}{23.6} \text{ MHz} = 3\text{ MHz} \quad (2''')$$

Simply calculation:

$$R'_L = \frac{R_L}{n_2^2} = 40\text{ k}\Omega \quad R'_s = R'_L \quad n_2 = \sqrt{\frac{R_s}{R'_L}} = \sqrt{\frac{10}{40}} = 0.5$$

$$R_0 \parallel R'_L \parallel R'_s = \frac{44.7 \times 20}{44.7 + 20} = 13.8\text{ k}\Omega$$

$$Q_L = \frac{R_0 \parallel R'_L \parallel R'_s}{\omega_0 L} = \frac{13.8 \times 10^3}{2\pi \times 71.2 \times 10^6 \times 10^{-6}} = 30.9 \quad B_{0.7} = \frac{f_0}{Q_L} = \frac{71.2}{30.9} \text{ MHz} = 2.3\text{ MHz}$$

2. (11') A Class-C resonant power amplifier with operating at critical point state. Assuming the output power $P_o=3W$, $E_c=13V$, the saturation voltage of transistor is $U_{ces}=1V$, the current conduction angle $\theta=60^\circ$, $\alpha_1(60^\circ)=0.218$, $\alpha_2(60^\circ)=0.391$.

- (1) Please calculate the DC power P_{DC} , power dissipation at the collector P_c , collector efficiency η_c .
- (2) In order to achieve the climate goal of carbon neutrality and emission peak, what measures can we take to improve the efficiency η_c of class-C power amplifier for RF energy saving? What do you think about the influences of index of high power for a power amplifier on wireless communication system, natural environment, and human health?

Solution:

$$(1) u_{c1m} = E_c - U_{ces} = (13-1)V = 12V$$

$$R_c = \frac{1}{2} \frac{u_{c1m}^2}{P_o} = \frac{12^2}{2 \times 3} \Omega = 24\Omega$$

$$I_{c1m} = \frac{u_{c1m}}{R_c} = \frac{12}{24} A = 500mA$$

$$I_{cm} = \frac{I_{c1m}}{\alpha_1(\theta)} = \frac{500}{0.391} mA = 1.28A$$

$$I_{c0} = I_{cm} \alpha_2(\theta) = 1.28 \times 0.218 A = 279mA$$

$$P_{DC} = E_c I_{c0} = 13 \times 279mW = 3.63W \quad (2')$$

$$P_c = P_{DC} - P_o = (3.63-3)W = 0.63W \quad (2')$$

$$\eta_c = \frac{P_o}{P_{DC}} = \frac{3}{3.63} \times 100\% = 82.6\% \quad (2')$$

- (2) Provide at least one correct way and explain the way for improving efficiency can get scores.(3')

$$\eta_c = \frac{1}{2} \frac{I_{c1m}^2}{I_{cm}^2} \frac{\alpha_1(\theta)}{\alpha_2(\theta)} \frac{u_{c1m}}{E_c}$$

Way-1: appropriately increase current conduction angle θ , usually ranges from $50^\circ \sim 70^\circ$.

Way-2: appropriately increase the collector voltage u_{c1m} , u_{c1m} decreases with the increase of u_{ces} and this change will lead to transistor working at saturation region.

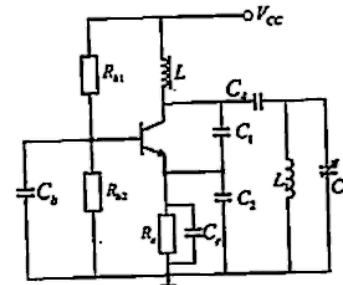
Way-3: appropriately increase the collector resistance R_c , based on the load characteristic of PA, the efficiency at critical point state isn't the maximum value. The maximum efficiency of class-C appears at the over-voltage status.

Understand about high power.(2')

At one hand, high power is helpful to realize long-distance signal transmission. At the other hand, higher power means stronger radiation power, which is harmful to environment and human body.

3. (10') The oscillation circuit is shown as below, assuming $V_{CC}=12V$, $R_{b1}=10k\Omega$, $R_{b2}=5k\Omega$, $R_e=1k\Omega$, $C_b=0.2\mu F$, $C_1=330pF$, $C_2=510pF$, $C_3=(2\sim30)pF$, $C_4=20pF$, $L_1=1\mu H$, $L=200\mu H$.

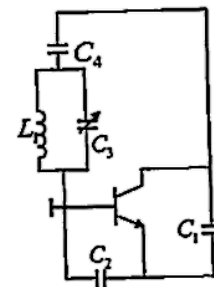
- (1) Draw the high frequency AC equivalent circuit and explain the type of oscillator.
- (2) Calculate the oscillating frequency ranges f_{min} , f_{max} and the feedback factor F .



Solution:

- (1) Seiler circuit or the second improved capacitor feedback LC oscillator.(2')

Draw correct equivalent circuit.(2')



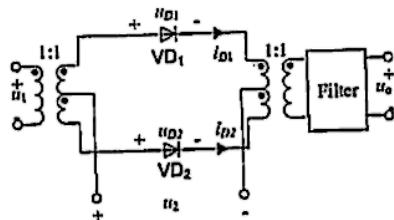
(2)

$$C_2 = C_1 + \frac{1}{\frac{1}{C_1} + \frac{1}{C_3}} = C_1 + C_3 = 22 \sim 50 pF \quad (2')$$

$$f_c = \frac{1}{2\pi \sqrt{L_1 C_2}} = \frac{1}{2\pi \sqrt{10^{-6} \times (22 \sim 50) \times 10^{-12}}} Hz = 22.5 \sim 33.9 MHz \quad (2')$$

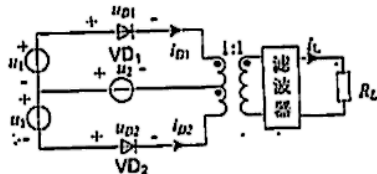
$$F = \frac{C_1}{C_1 + C_2} = \frac{330}{330 + 510} = 0.39 \quad (2')$$

4. (20') The diode multiplier circuit is shown as below. Assume that the diodes VD1 and VD2 are ideal component and exactly the same. $u_1(t) = u_{1m} \cos \Omega t \cos \omega_L t$, $u_2(t) = u_{2m} \cos \omega_L t$, $u_{2m} \gg u_{1m}$, the conductance of diodes is g_D .



- (1) If the equivalent resistance of the filter at the center frequency f_0 is R_L , write the expression of output voltage $u_o(t)$ and list the frequency component of $u_o(t)$.
- (2) Please draw the frequency spectra of $u_1(t)$, $u_2(t)$, and $u_o(t)$.
- (3) What type of filter should be used at u_o terminal? Please write the center frequency f_0 and the bandwidth B of the filter.

Solution:



$$\begin{aligned}
 (1) \quad & u_{D1} = u_1 + u_2 \dots\dots\dots(1') \\
 & u_{D2} = -u_1 + u_2 \dots\dots\dots(1') \\
 & i_{D1} = g_D K_1(\omega_L t)(u_1 + u_2) \dots\dots\dots(1') \\
 & i_{D2} = g_D K_1(\omega_L t)(-u_1 + u_2) \dots\dots\dots(1') \\
 & u_1 = u_{1m} \cos \Omega t \cos \omega_L t = \frac{1}{2} u_{1m} [\cos(\omega_L - \Omega)t + \cos(\omega_L + \Omega)t] \dots\dots\dots(1') \\
 & K_1(\omega_L t) = \frac{1}{2} + \frac{2}{\pi} \cos \omega_L t - \frac{2}{3\pi} \cos 3\omega_L t + \frac{2}{5\pi} \cos 5\omega_L t - \dots\dots\dots(2')
 \end{aligned}$$

$$\begin{aligned}
 i_{D1} - i_{D2} &= 2g_D K_1(\omega_L t) \cdot u_1 \\
 &= g_D u_{1m} [\cos(\omega_L - \Omega)t + \cos(\omega_L + \Omega)t] \left[\frac{1}{2} + \frac{2}{\pi} \cos \omega_L t - \frac{2}{3\pi} \cos 3\omega_L t + \frac{2}{5\pi} \cos 5\omega_L t - \dots \right] \dots\dots\dots(2')
 \end{aligned}$$

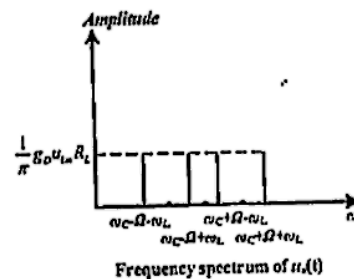
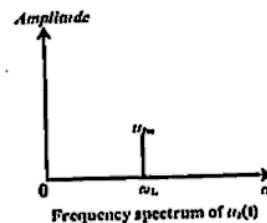
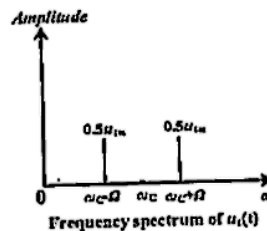
$$\begin{aligned}
 u_o(t) &= i_L R_L = \frac{2}{\pi} g_D u_{1m} R_L [\cos(\omega_L - \Omega)t + \cos(\omega_L + \Omega)t] \cos \omega_L t \\
 &= \frac{1}{\pi} g_D u_{1m} R_L [\cos(\omega_L - \Omega - \omega_L)t + \cos(\omega_L - \Omega + \omega_L)t + \cos(\omega_L + \Omega - \omega_L)t + \cos(\omega_L + \Omega + \omega_L)t] \dots\dots\dots(2')
 \end{aligned}$$

The frequency component of $u_o(t)$ is $\omega_c - \Omega \pm \omega_L$, and $\omega_c + \Omega \pm \omega_L$(1')

(2) Frequency spectrum of $u_1(t)$,(2')

Frequency spectrum of $u_2(t)$,(1')

Frequency spectrum of $u_o(t)$ (2')



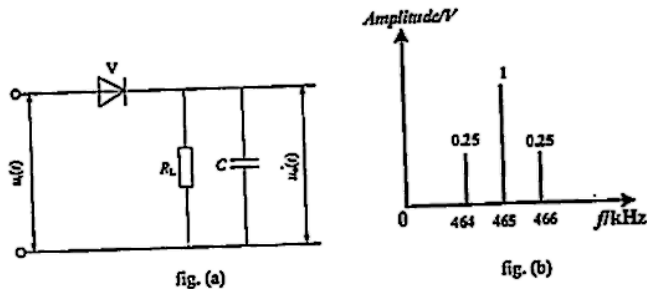
(3) Bandpass filter.(1')

$$f_0 = \frac{\omega_c}{2\pi} \dots\dots\dots(1')$$

$$B = \frac{2(\Omega + \omega_L)}{2\pi} = \frac{\Omega + \omega_L}{\pi} \dots\dots\dots(1')$$

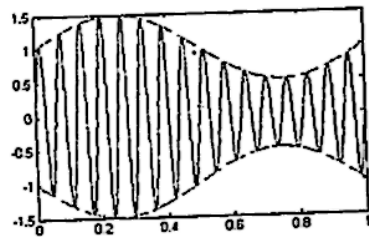
5. (14') The circuit is shown as below in fig. (a), assuming $C=0.05\mu F$, the frequency spectrum of input signal $u(t)$ is given in fig. (b).

- (1) Please write the mathematical expression of the input signal $u(t)$, draw the time domain waveform and explain what kind of modulation signal is it?
- (2) For fig. (a), what kind of circuit is it? What is the frequency f_c of the output signal $u_o(t)$?
- (3) How to choose the load resistance R_L to prevent diagonal distortion? How to choose the load resistance R_L to acquire high detection efficiency?



Solution:

- (1) $u(t) = \cos 2\pi \times 465 \times 10^3 t + 0.25 \cos 2\pi \times 466 \times 10^3 t + 0.25 \cos 2\pi \times 464 \times 10^3 t$ (v)
 $= \cos 2\pi \times 465 \times 10^3 t (1 + 0.5 \cos 2\pi \times 10^3 t)$ (v).....(2')
 Standard AM wave or AM wave.....(2')



Time domain waveform.....(2')

- (2) Large signal peak envelope detector.....(2')
 $f_c = \frac{\Omega}{2\pi} = 1 \text{ KHz}$(2')
- (3)

$$m_a = 0.5$$

$$\Omega = 2\pi \times 10^3 \text{ rad/s}$$

$$R_L < \frac{\sqrt{1-m_a^2}}{m_a} \times \frac{1}{\Omega C} = \frac{\sqrt{0.75}}{0.5} \times \frac{1}{2\pi \times 10^3 \times 0.05 \times 10^{-6}} \Omega = 5.5 \text{ K}\Omega \dots\dots\dots(2')$$

$$\omega_c = 2\pi \times 465 \times 10^3 \text{ rad/s}$$

$$R_L \gg \frac{1}{\omega_c C} = \frac{1}{2\pi \times 465 \times 10^3 \times 0.05 \times 10^{-6}} \Omega = 6.8 \Omega \dots\dots\dots(2')$$

6. (5') The phase modulated signal $u_{PM}(t) = 10 \cos(2\pi \times 10^6 t + 10 \cos 2\pi \times 10^3 t)$ (v).

- (1) Find out the carrier wave frequency f_c , modulation signal frequency F , and PM factor m_p ?
- (2) Calculate the maximum frequency shift Δf and the effective bandwidth B of the PM signal.

Solution:

$$(1) u_{PM}(t) = 10 \cos(2\pi \times 10^6 t + 10 \cos 2\pi \times 10^3 t) = U_m \cos(\omega_c t + m_p \cos \Omega t)$$

$$f_c = \frac{\omega_c}{2\pi} = 1 \text{ MHz} \dots\dots\dots(1')$$

$$F = \frac{\Omega}{2\pi} = 1 \text{ KHz} \dots\dots\dots(1')$$

$$m_p = 10 \dots\dots\dots(1')$$

(2)

$$\Delta f_p = \frac{\Delta \omega_p}{2\pi} = \frac{m_p \Omega}{2\pi} = \frac{10 \times 2\pi \times 10^3}{2\pi} \text{ Hz} = 10 \text{ KHz} \dots\dots\dots(1')$$

$$B = 2(m_p + 1)F = 2(\Delta f_p + F) = 2(10 + 1) \text{ KHz} = 22 \text{ KHz} \dots\dots\dots(1')$$